

# **Prototype 3**

## **Plan for 4**

Palmeiras

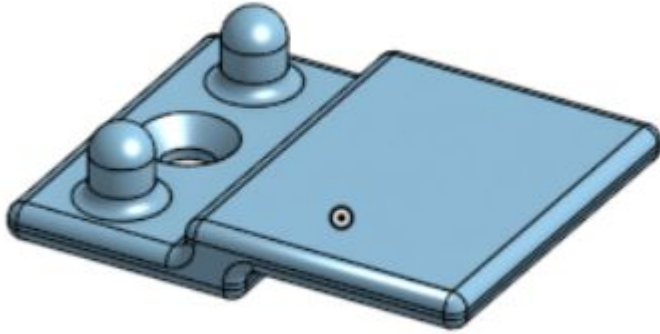
# Attempted Prototype:

- Homemade window
- Detailing of water system, exploration of different ideas
- Motor control full coding for length of window
- Mounted System on Window

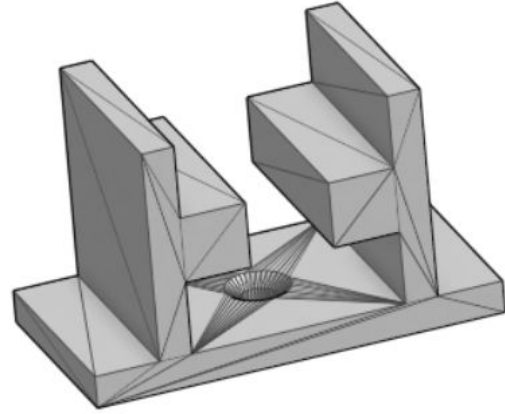
## Successes of Prototype 3

- Gathered Supplies and found cost effective solutions to the window, power supply
- Met with Professor Marks to talk about the water system, came up with two ideas, told to make thorough schematic and material list before meeting, so decision could be made in conjunction with TA whether to purchase
- Formulated working Arduino code to span full length of mirror on command
- Made effective designs, and printed, to mount the glass on wooden frame and the T-slot design to the wooden frame

# Prototype 3 Products



Window Clip Design



T-slot to window  
mount

# Pitfalls of Prototype 3

- Were unable to build window frame until late
  - Did not have access to strong enough drill for screws needed
  - Could not get makerspace clamps to work on glue for wooden frame. Frame was too small
- This led to us not having anything to mount the prototype to, nor to test on the window the minor details of the code
- Barrel Jack adapter, limit switches not going to arrive in time
  - Didn't have access to power supply between meetings

# Fixing Shortcomings of Prototype 3

- Re-printing fasteners for window to window frame
- Got smaller screws to build frame

# Prototype 4

- Must be able to spray water across the window
- Has to have a full length squeegee spanning the height of the window
- Must be an applied force on the squeegee to improve cleaning

Window Dimensions:

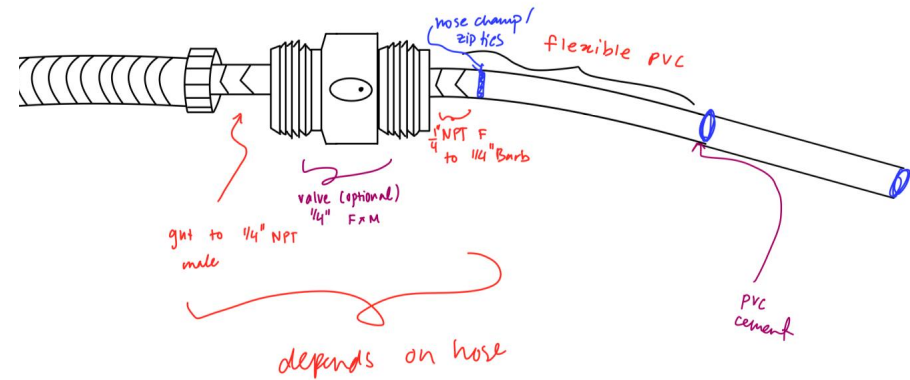
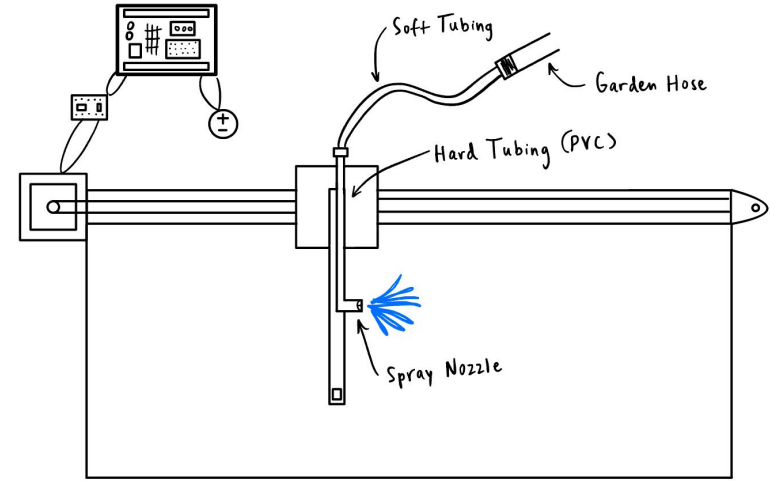
19 x 15 inches (of exposed glass)

→ Squeegee = 15 inches long

→ x-range = 19 inches

# Prototype 4

- Plan to add a water dispensing system
- Will be attached to garden hose and spray water on the window before wiping motion
- Ideally would spray water on same schedule as the squeegee





# Water System Items

Fittings:

[Garden Hose to Adapter \(GHT x 1/4" NPT Male\)](#)

[Valve \(1/4" NPT Female Male, Ball Valve\)](#)

[Barbed Adapter \(1/4" NPT Female x 1/4" Barb\)](#)

[Polyethylene Tubing](#)

- 1/4 ID and 3/8 OD
- ~160 psi (not rated on website, compared to similar items)
- Bend radius ~ 2 inches
- McMaster-Carr has PVC alternative
- Need Barbed connection for Polyethylene, not for PVC (cement)

Type of connection from hose to pvc depends on the type of hose we have available, as well as whether we want the valve or not (likely not, may be too much out of budget)

Can just use [GHT x Barb](#), straight into flexible PVC/Polyethylene

► About Plastic and Rubber Tubing  
More

### Soft Plastic Tubing for Air and Water

This general purpose PVC tubing is the industry standard for air and water applications. Tubing is clear, so you can easily monitor flow. It is flexible to easily bend around objects.

**Masterkleer** tubing is more economical than Tygon tubing. **Tygon** tubing has a nonabsorbent surface for improved flow and drainage.

For technical drawings and 3-D models, click on a part number.

**Clear Masterkleer PVC Plastic**

- Flexible
- Compatible with Barbed Tube Fittings
- Hardness: Soft (Durometer 70A)
- Temperature Range: -40° to 140° F

| ID   | OD    | Wall Thick. | Bend Radius | Max. Pressure  | Color | Lengths, ft. | Per Ft.         |
|------|-------|-------------|-------------|----------------|-------|--------------|-----------------|
| 1/4" | 5/16" | 1/32"       | 1 1/4"      | 10 psi @ 72° F | Clear | 50, 100      | 5233K947 \$0.81 |
| 1/2" | 3/8"  | 1/16"       | 1"          | 35 psi @ 72° F | Clear | 25, 50, 100  | 5233K56 .42     |

Soft Masterkleer PVC Tubing for Air and Water, 1/4" ID, 3/8" OD

Length, ft.  
✓ 25

ADD TO ORDER  
Delivers tomorrow  
\$10.50 Each

Product Detail CAD 3-D Solidworks Download

### Flat Spray Nozzle

Brass, 1/4 NPT Male, 1.0 gpm At 20 PSI

Spray Angle  
✓ 120°

1 Each  
ADD TO ORDER  
1-9 Each \$8.96  
10 or more \$8.07  
Delivers tomorrow  
3404K14

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**Solidworks Add-In**  
Available for Solidworks 2017 or newer.

Connection Type Pipe  
Bore Connection Type Threaded

# Calculation

$$v = \frac{Q}{A}$$

$$D = 0.00635 \text{ m}$$

$$v = 1.05 \text{ m/s}$$

$$Q = 2 \frac{\text{L}}{\text{min}}$$

$$Re = \frac{1000 \cdot 1.05 \cdot 0.00635}{0.001} = 6680$$

$$f \approx \underbrace{0.035}_{\text{Moody chart}}$$

$$h_f = f \cdot \frac{L}{D} \cdot \frac{v^2}{2g}$$

$$h_f = 62 \text{ cm}$$

$$\begin{aligned} & \underbrace{62 \text{ cm}} + \underbrace{100 \text{ cm}}_{\substack{\text{assumed} \\ \text{elevation}}} + \underbrace{12 \text{ cm}}_{\text{minor}} \\ &= 174 \text{ cm} \quad \text{head loss} \end{aligned}$$

# Timeline

- Week 7:
  - Decide on water system
    - Order parts for system (potentially)
  - Power motor directly from an outlet
  - Mount device on window frame
  - Squeegee arm
- Week 8:
  - Build water system
  - Plan and build housing
- Week 9:
  - Finish housing
  - Polish it up

# Division of Work

- Lucky: worked on making window and mounts for T-slot
- Paul: worked on writing code for motor/limit switches
- Neha: planned next prototype
- Flynn: worked on making window and clasps for window frame